

50.007 Machine Learning

Logistic Regression

Roy Ka-Wei Lee
Assistant Professor, ISTD, SUTD



Recap: Linear SVM

- High-level idea of SVM: Finding the most optimal hyperplane by maximizing the margin of S!

Original Objective Function (Primal)

Primal problem

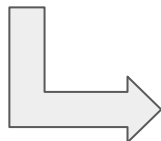
Given data set : $S = \{(\mathbf{x}_i, d_i)\}, i = 1, 2, \dots, N$

Find : $\mathbf{w}$ and b

Minimizing : $f(\mathbf{w}) = \frac{1}{2} \mathbf{w}^T \mathbf{w}$

Subject to : $d_i (\mathbf{w}^T \mathbf{x}_i + b) \geq 1$

Applying
Lagrange
Theorem...



Dual Problem

Finding optimal hyperplane (dual problem)

Given : $S = \{(\mathbf{x}_i, d_i)\}$

Find : Lagrange multipliers $\{\alpha_i\}$

Maximizing : $Q(\alpha) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j d_i d_j \mathbf{x}_i^T \mathbf{x}_j$

Subject to : (1) $\sum_{i=1}^N \alpha_i d_i = 0$

(2) $\alpha_i \geq 0$

Apply **KKT Conditions**,
solve α to find $\mathbf{w}$ and b

$$\frac{\partial L(\mathbf{w}, b, \alpha)}{\partial \mathbf{w}} = 0$$

$$\frac{\partial L(\mathbf{w}, b, \alpha)}{\partial b} = 0$$

$$d_i (\mathbf{w}^T \mathbf{x}_i + b) \geq 1$$

$$\alpha_i (d_i (\mathbf{w}^T \mathbf{x}_i + b) - 1) = 0$$

$$\alpha_i \geq 0$$

$$\mathbf{w}_o = \sum_{i=1}^N \alpha_{o,i} d_i \mathbf{x}_i, \quad b_o = \frac{1}{d^{(s)}} - \mathbf{w}_o^T \mathbf{x}^{(s)}$$

Recap: Non-Linear SVM - Soft Margin

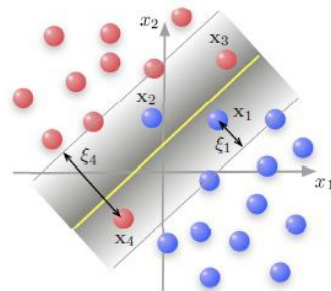
- Convert (classification) **hard constraints** to **soft constraints**
- Introduce a Nonnegative **slack variables** $\xi_i \geq 0$
 - Data point not in region of separation $\xi_i = 0$
 - Data point in region of separation and on correct side of hyperplane $0 \leq \xi_i \leq 1$ (ξ_1, ξ_3)
 - Data point in region of separation but on wrong side of hyperplane $\xi_i > 1$ (ξ_3, ξ_4, ξ_1)

Minimize: $\frac{1}{2} w^T w + C \sum_{i=1}^N \xi_i$

Subject to:

(i) $d_i(w^T x_i + b) - 1 + \xi_i \geq 0$

(ii) $\xi_i \geq 0$



A **smaller C value** will result in a **larger margin** with more tolerance for misclassifications

A **larger C value** will result in a **smaller margin** with less tolerance for misclassifications

Recap: Non-Linear SVM - Kernel

- Map data to a **higher dimensional** space, do linear classification there.

Dual problem with soft margin and transformation

Find : α_i

$$\text{Maximize : } Q(\alpha) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j d_i d_j \boxed{\varphi^T(\mathbf{x}_i) \varphi(\mathbf{x}_j)} \longleftrightarrow K(\mathbf{x}_i, \mathbf{x}_j)$$

Kernel trick

$$\text{Subject to : } \sum_{i=1}^N \alpha_i d_i = 0, \quad 0 \leq \alpha_i \leq C$$

- Use **Kernel trick**, which give an *efficient* way (*kernel function*) to compute dot products in some feature space without even knowing what this space is and what is $\phi(x)$.
- We can build many common kernel functions based only on **four rules** (see previous slides and lecture notes)

Learning Objectives

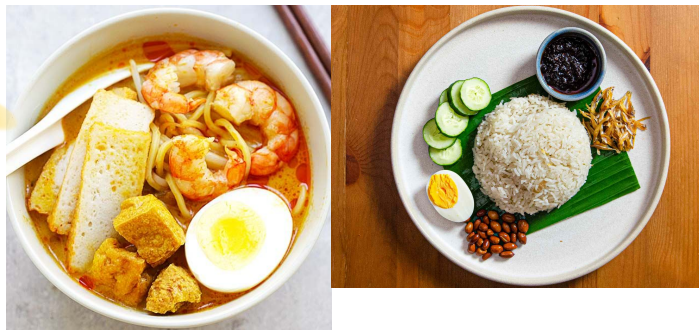
You should be able to know:

- What is Logistic Regression? What type of problems logistic regression can solve?
- What does the decision boundary of logistic regression look like?
- How to perform learning (*i.e.* training) for logistic regression?
- How to perform prediction (*i.e.* inference) under logistic regression?

What is Logistic Regression?

- A discriminative classifier.
- Logistic regression can be used to classify an observation into one of two classes (like 'positive sentiment' and 'negative sentiment'), or into one of many classes.
- In this lecture, we'll study two-class case.

A **generative** model would have the goal of understanding what Laksa look like and what cats look like. You might literally ask such a model to 'generate', i.e. draw laksa.



A **discriminative** model, by contrast, is only trying to learn to distinguish the classes (perhaps without learning much about them)

What is Classification?

- It is a **probabilistic** classifier that makes use of **supervised** machine learning.
- Machine learning classifiers require a training corpus of **M input/output** pairs (x^i, y^i) , where superscript **i** refer to the **specific data** instance.
- For each input observation x^i , this will be a vector of features $\begin{bmatrix} x_0 \\ x_1 \\ \dots \\ x_n \end{bmatrix}$
- We will generally refer to feature **j** of input x^i as x_j^i or simply x_j .

Note that in certain material,
the notation may change.
Read closely!



Big Picture of Logistic Regression

Comparing to what you have learned so far...

Comparing to what you have learned

- Support Vector Machine
- Linear Regression

Why I still need to learn
Logistic Regression when
I we just barely survived
SVM?! Isn't both doing
classification?!



Comparing to what you have learned

- Support Vector Machine 
- Linear Regression

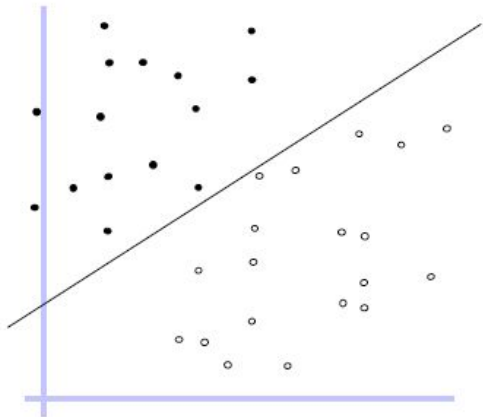
Why I still need to learn
Logistic Regression when
I we just barely survived
SVM?! Isn't both doing
classification?!



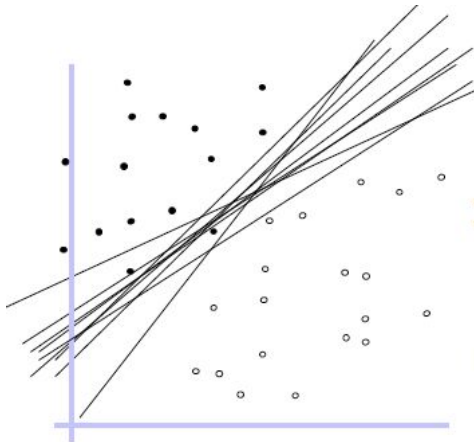
SVM (Recall)

Linearly Separable

'Decision boundary'



Any of these would be fine...



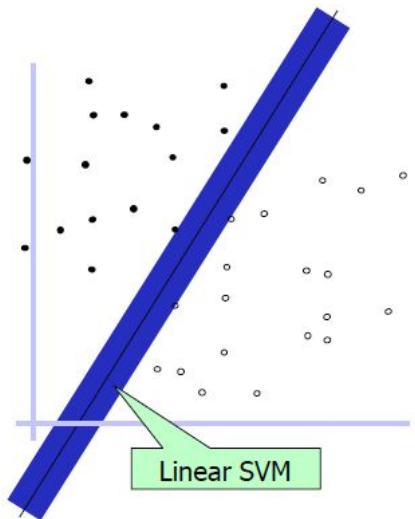
which decision
boundary is the
best?



SVM (Recall)

Linearly Separable

Optimal hyperplane: the one that maximizes the margin



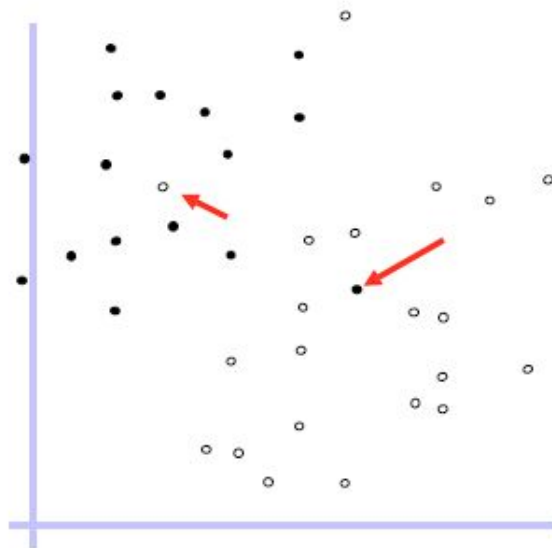
Margin: The width that the boundary can be increased before hitting a data point.



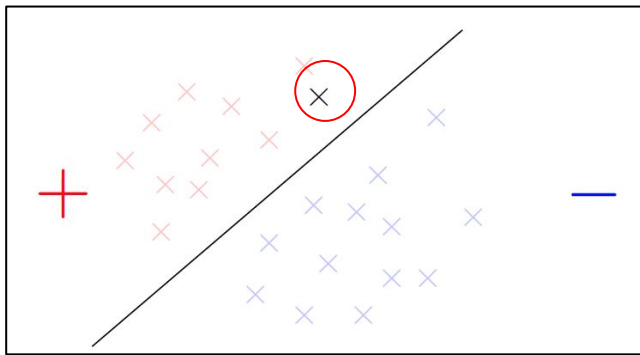
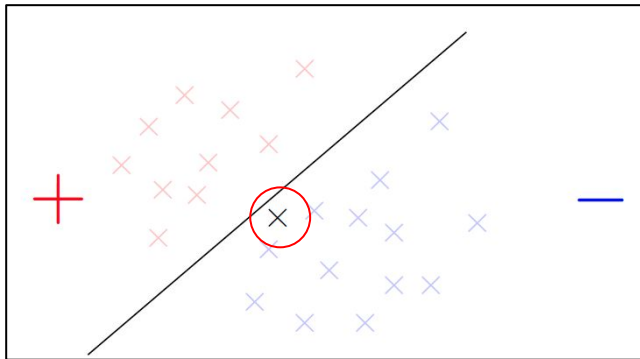
SVM: not linearly separable data (Recall)

Two solutions:

1. Soft margin: Allow a few points on the wrong side (slack variables), and/or
2. Kernel trick: Map data to a higher dimensional space, do linear classification there.



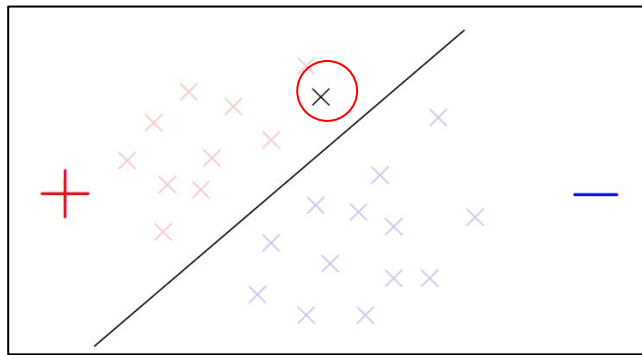
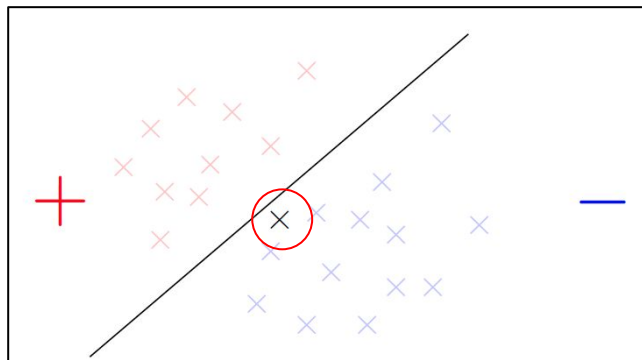
Classifier Evaluation



Let's assume that we are done with training of SVM,
What should be the label of these points?



Classifier Evaluation

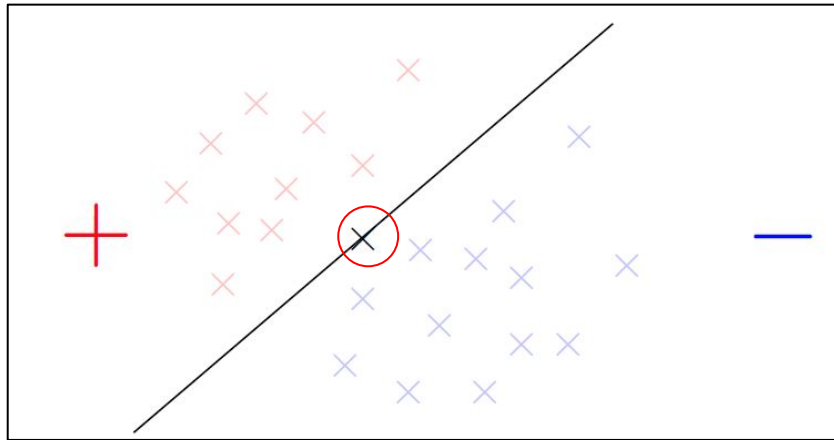


+

Let's assume that we are done with training of SVM,
What should be the label of these points?



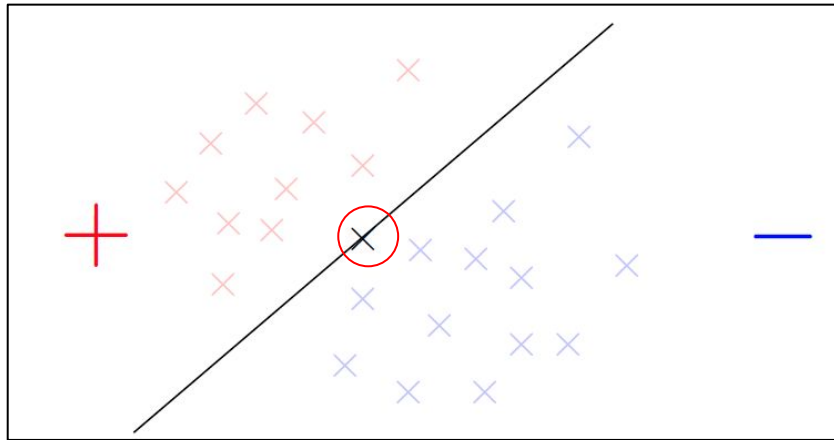
Classifier Evaluation



How about this point?



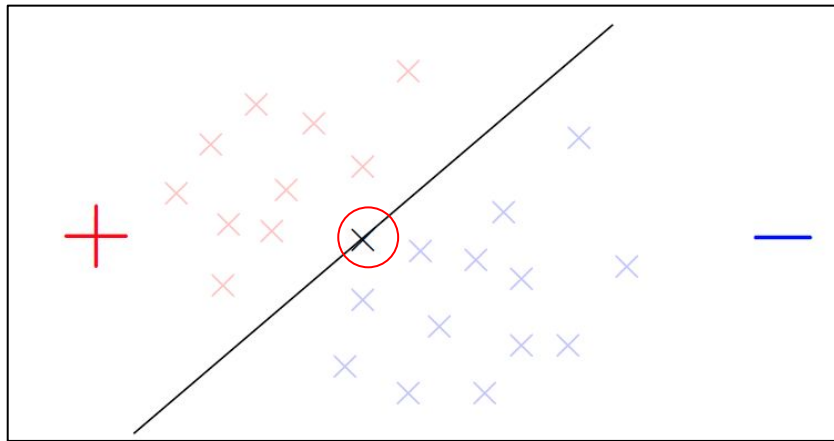
Classifier Evaluation



Is it possible to introduce
the notion of
confidence/probability
score into the model?



Classifier Evaluation



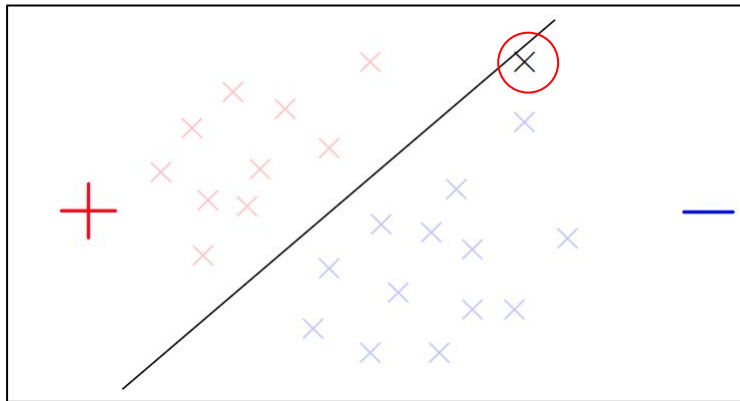
50% positive, 50% negative

Logistic Regression: “Linear classification:
classification with probability”

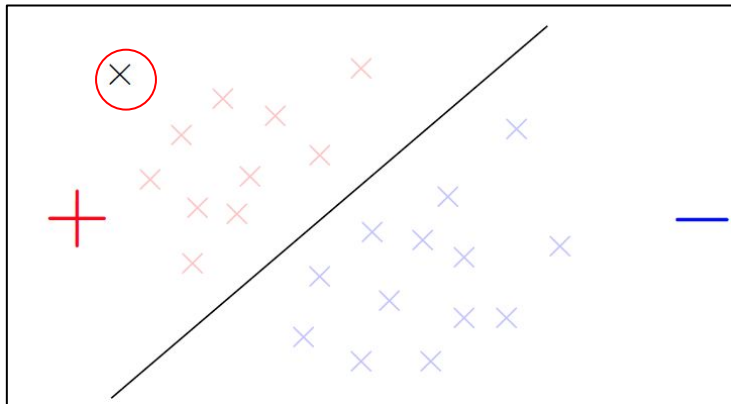
Is it possible to introduce
the notion of
confidence/probability
score into the model?



Classifier Evaluation



45% positive, 55% negative



80% positive, 20% negative

Logistic Regression: “Linear classification:
classification with probability”

Classification with Probabilistic Output

- In some binary classification tasks, a **probabilistic output** would be *appropriate* than a **binary decision**.
 - Spam email classification: spam or not spam?
 - Credit risk assessment: default on their loan or not?
 - Fraud detection: fraudulent or Not?
 - Tumor diagnosis: malignant or benign?

$$y \in \{0, 1\}$$

0: "Negative Class" (e.g., benign tumor)

1: "Positive Class" (e.g., malignant tumor)

Comparing to what you have learned

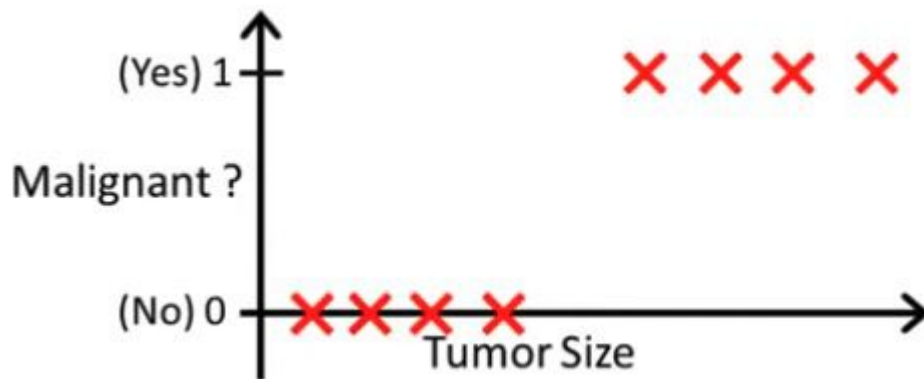
- Support Vector Machine
- Linear Regression



Classification with Linear Regression

$y \in \{0,1\}$:

0: “negative class” (benign tumor), 1: “positive class” (malignant tumor)



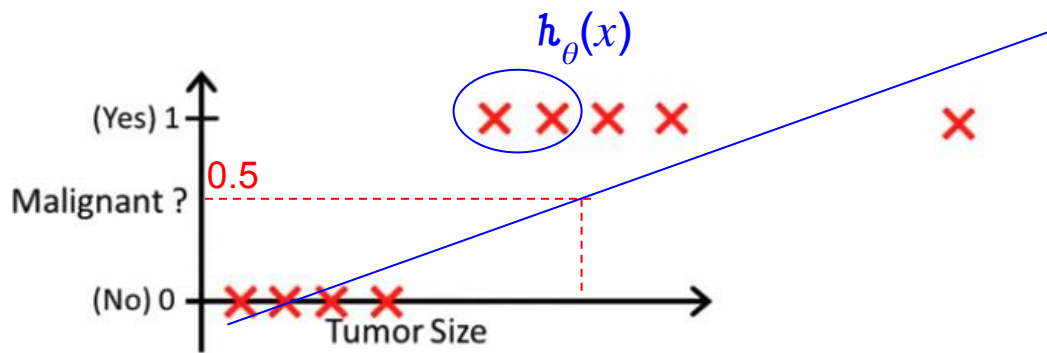
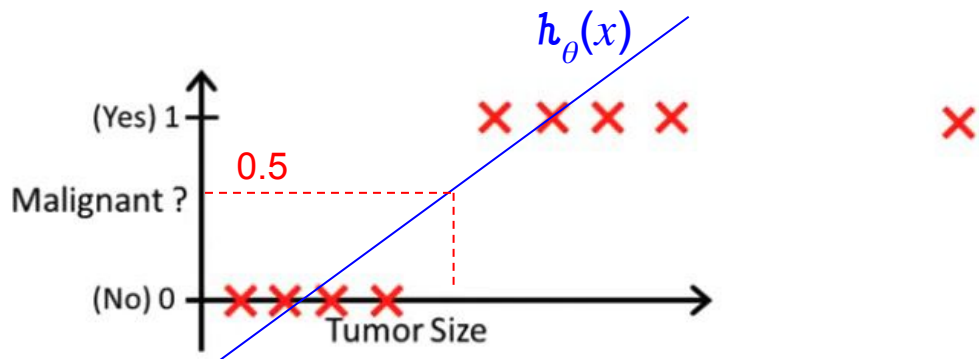
How will you
classify this using
Linear Regression



Classification with Linear Regression

Let's add one more point to the training data...

If we run linear regression with the new data:

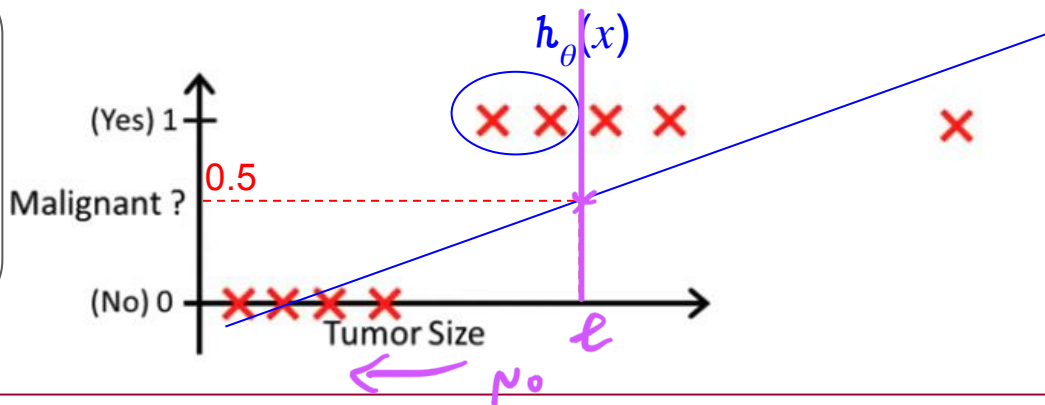
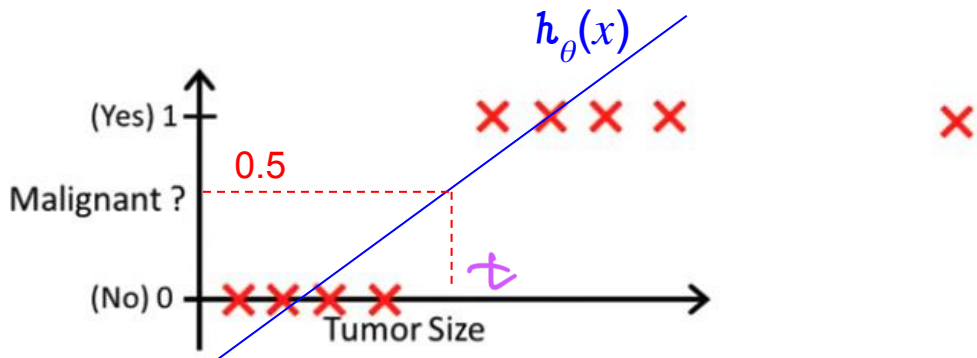


Classification with Linear Regression

Let's add one more point to the training data...

If we run linear regression with the new data:

It is a bad classification... By adding one example, we can decrease the accuracy a lot. So, linear regression is often not a good classification method. Previously, linear regression was lucky!



Logistic Regression

Intuition & Basic Definition...

Logistic Regression as a Classification Algorithm

We would like to have a classifier that outputs values between 0 and 1:

$$0 \leq h_{\theta}(x) \leq 1$$

(classification: $y = 0$ or $y = 1$)

Linear regression: $h_{\theta}(x) = \theta^T x$

Logistic regression: $h_{\theta}(x) = g(\theta^T x)$

Where:

$$g(z) = \frac{1}{1 + e^{-z}}$$

Hypothesis function

$$h(\theta) = \frac{1}{1 + e^{-\theta^T x}}$$

Sigmoid function =
logistic function.

But why
sigmoid/logistic
function?



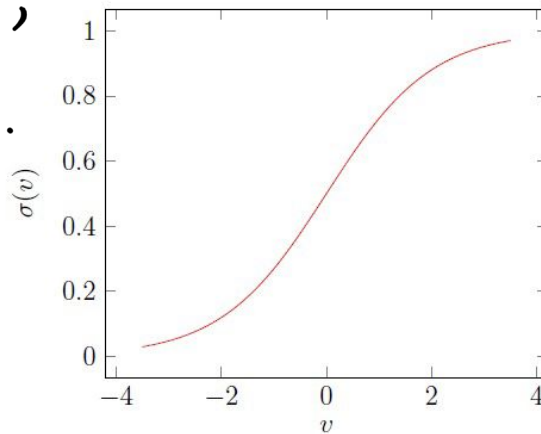
Logistic Regression

Why sigmoid/logistic function?

1. Maps any number in between 0 and 1.
2. Interpret the results as a probability

The cost function is constructed to maximize the probability of correct classification. (Easy to work with!)

As matter if the #
is the or -ve,
it maps to
a value b/w.
0 & 2.



How do you interpret this model?



Logistic Regression

Interpretation

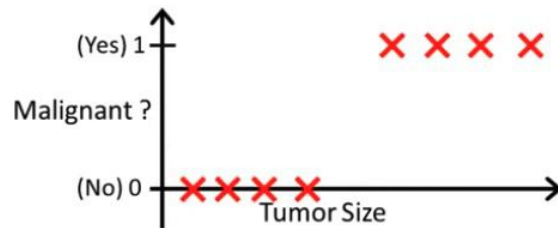
$h_{\theta}(x)$ = the estimated probability that $y = 1$ on input x . Let's see an example together.

$y \in \{0,1\}$:

0: "negative class" (benign tumor),

1: "positive class" (malignant tumor)

x : tumor size



- $h_{\theta}(x) = 0.7 \rightarrow$ tell the patient that 70% probability for tumour to be malignant :’(
- $h_{\theta}(x) = p(y = 1 | x; \theta)$ “probability that $y = 1$, given x , parameterized by θ ”
- Note that y can only be 0 or 1 $\rightarrow p(y = 1 | x; \theta) + p(y = 0 | x; \theta) = 1$



Logistic Regression - Decision Boundary

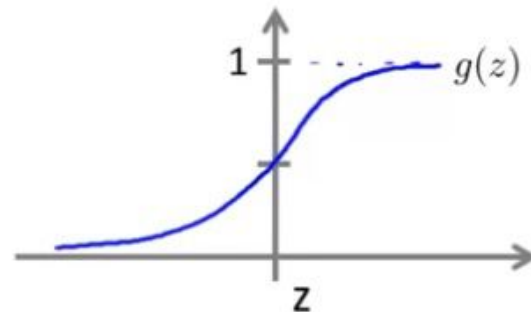
Hypothesis Function

$$h_{\theta}(x) = g(\theta^T x)$$

$$\theta^T x = z$$

$$g(z) = \frac{1}{1+e^{-z}}$$

$$h_{\theta}(x) = p(y = 1 | x, \theta)$$



$$g(z) \geq 0.5 \text{ if } z \geq 0$$

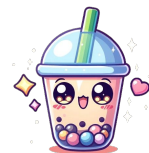
$$h_{\theta}(x) = g(\theta^T x) \geq 0.5$$

$$\text{whenever } z = \theta^T x \geq 0$$

Threshold the classifier output at 0.5

$$h_{\theta}(x) \geq 0.5 \text{ predict } y = 1 \quad \rightarrow \quad \theta^T x \geq 0$$

$$h_{\theta}(x) < 0.5 \text{ predict } y = 0 \quad \rightarrow \quad \theta^T x < 0$$



Logistic Regression - Decision Boundary

Example 1: Linear decision boundary

Let us assume our hypothesis function is:

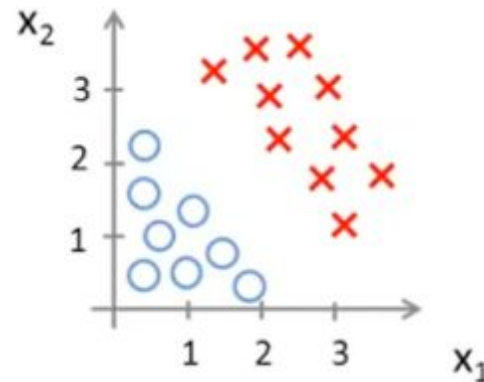
$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

When we set the parameters as: $\theta = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$

We predict

$$y = 1 \text{ if } -3 + x_1 + x_2 \geq 0$$

$$y = 0 \text{ if } -3 + x_1 + x_2 < 0$$



Logistic Regression - Decision Boundary

Example 1: Linear decision boundary

Let us assume our hypothesis function is:

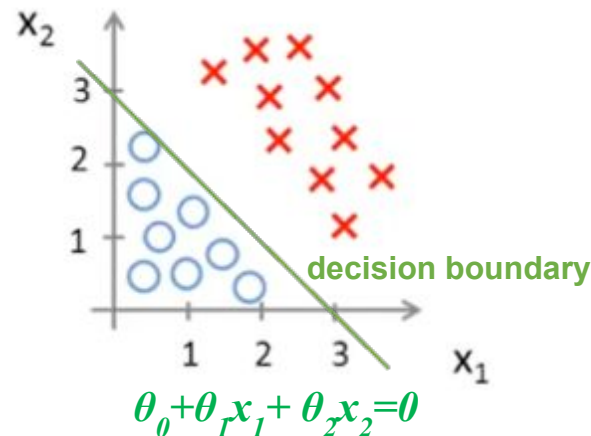
$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

When we set the parameters as: $\theta = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$

We predict

$$y = 1 \text{ if } -3 + x_1 + x_2 \geq 0$$

$$y = 0 \text{ if } -3 + x_1 + x_2 < 0$$



Logistic Regression - Decision Boundary

Example 1: Linear decision boundary

Let us assume our hypothesis function is:

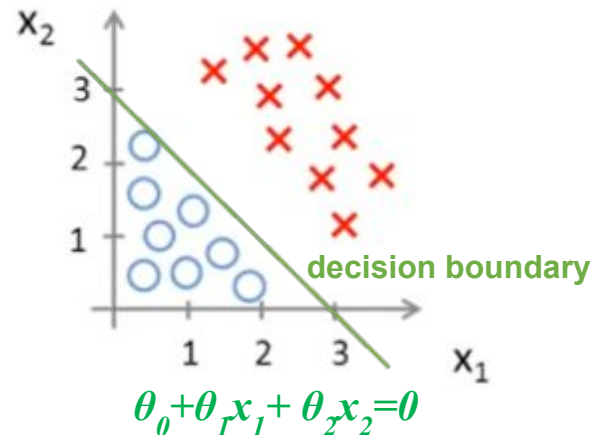
$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

When we set the parameters as: $\theta = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$

We predict

$$y = 1 \text{ if } -3 + x_1 + x_2 \geq 0$$

$$y = 0 \text{ if } -3 + x_1 + x_2 < 0$$



Erm... So what does this have to do with logistic regression?



Logistic Regression - Decision Boundary

Example 1: Linear decision boundary

Suppose we have data point where $(x_1, x_2) = (1, 1)$, what's its y ?

Recall

$$\begin{aligned}h_{\theta}(x) &= g(\theta^T x) \\ \theta^T x &= z \\ g(z) &= \frac{1}{1+e^{-z}} \\ h_{\theta}(x) &= p(y = 1|x, \theta)\end{aligned}$$

$$y = 1 \text{ if } -3 + x_1 + x_2 \geq 0$$

$$y = 0 \text{ if } -3 + x_1 + x_2 < 0$$

y = 0 because $-3 + 1 + 1 < 0$

$$z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 = -3 + 1 + 1$$

$$g(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^1} = 0.269$$

$$h_{\theta}(x) \geq 0.5 \text{ predict } y = 1 \quad \rightarrow \quad \theta^T x \geq 0$$

$$h_{\theta}(x) < 0.5 \text{ predict } y = 0 \quad \rightarrow \quad \theta^T x < 0$$

Since $\theta^T x = z$

$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2$

$= -3 + 1(1) + 1(1) = -1$

$g(\theta^T x) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-(-1)}} = \frac{1}{1+e^1} = 0.269$

$P(y=1 | x; \theta) = 0.269$

Logistic Regression - Decision Boundary

Example 1: Linear decision boundary

Suppose we have data point where $(x_1, x_2) = (1, 1)$, what's its y ?

$$y = 1 \text{ if } -3 + x_1 + x_2 \geq 0$$

$$y = 0 \text{ if } -3 + x_1 + x_2 < 0$$

$y = 0$ because $-3 + 1 + 1 < 0$

$$z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 = -3 + 1 + 1$$

$$g(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^1} = 0.269$$

$$h_{\theta}(x) \geq 0.5 \text{ predict } y = 1 \quad \rightarrow \quad \theta^T x \geq 0$$

$$h_{\theta}(x) < 0.5 \text{ predict } y = 0 \quad \rightarrow \quad \theta^T x < 0$$

Recall

$$h_{\theta}(x) = g(\theta^T x)$$

$$\theta^T x = z$$

$$g(z) = \frac{1}{1+e^{-z}}$$

$$h_{\theta}(x) = p(y = 1 | x, \theta)$$

What does 0.269 mean?



Logistic Regression - Decision Boundary

Example 1: Linear decision boundary

Suppose we have data point where $(x_1, x_2) = (1, 1)$, what's its y ?

$$y = 1 \text{ if } -3 + x_1 + x_2 \geq 0$$

$$y = 0 \text{ if } -3 + x_1 + x_2 < 0$$

$y = 0$ because $-3 + 1 + 1 < 0$

$$z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 = -3 + 1 + 1$$

$$g(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^1} = 0.269$$

$$h_{\theta}(x) \geq 0.5 \text{ predict } y = 1 \quad \rightarrow \quad \theta^T x \geq 0$$

$$h_{\theta}(x) < 0.5 \text{ predict } y = 0 \quad \rightarrow \quad \theta^T x < 0$$

Recall

$$h_{\theta}(x) = g(\theta^T x)$$

$$\theta^T x = z$$

$$g(z) = \frac{1}{1+e^{-z}}$$

$$h_{\theta}(x) = p(y = 1 | x, \theta)$$

It means it has a probability of 0.269 being $y=1$ or a 0.731 chance of being $y=0$

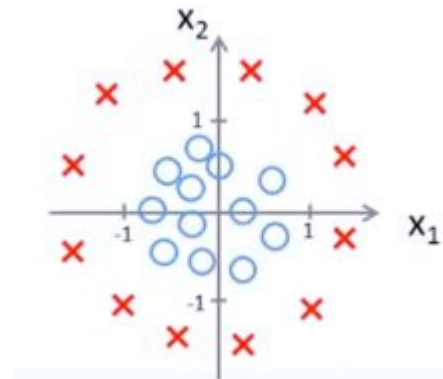


Logistic Regression - Decision Boundary

Example 2: Non-Linear decision boundary

Let us assume our hypothesis function is:

$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2) \text{ and } \theta = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$



We predict

$$y = 1 \text{ if } -1 + x_1^2 + x_2^2 \geq 0$$

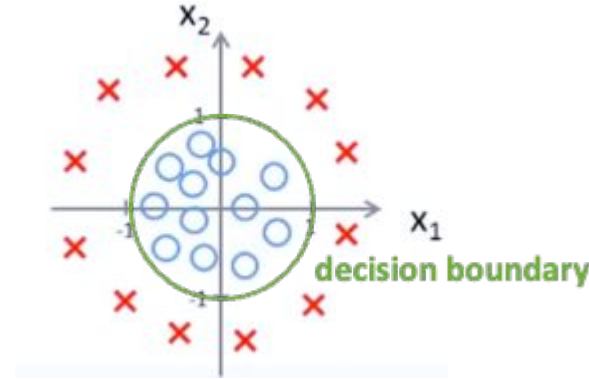
$$y = 0 \text{ if } -1 + x_1^2 + x_2^2 < 0$$

Logistic Regression - Decision Boundary

Example 2: Non-Linear decision boundary

Let us assume our hypothesis function is:

$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2) \text{ and } \theta = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$



We predict

$$\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2 = 0$$

$$y = 1 \text{ if } -1 + x_1^2 + x_2^2 \geq 0$$

$$y = 0 \text{ if } -1 + x_1^2 + x_2^2 < 0$$

Logistic Regression - Decision Boundary

Example 2: Non-Linear decision boundary

Suppose we have data point where $(x_1, x_2) = (2, 2)$, what's its y ?

$$y = 1 \text{ if } -1 + x_1^2 + x_2^2 \geq 0$$

$$y = 0 \text{ if } -1 + x_1^2 + x_2^2 < 0$$

$$y = 1 \text{ because } -1 + 2^2 + 2^2 \geq 0$$

$$z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2 = -1 + 0 + 0 + 4 + 4 = 7$$

$$g(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-7}} = 0.999$$

$$h_{\theta}(x) \geq 0.5 \text{ predict } y = 1 \quad \rightarrow \quad \theta^T x \geq 0$$

$$h_{\theta}(x) < 0.5 \text{ predict } y = 0 \quad \rightarrow \quad \theta^T x < 0$$

Recall

$$h_{\theta}(x) = g(\theta^T x)$$

$$\theta^T x = z$$

$$g(z) = \frac{1}{1+e^{-z}}$$

$$h_{\theta}(x) = p(y = 1 | x, \theta)$$

It means it has a probability of 0.999 being $y=1$ or a 0.0001 chance of being $y=0$



Logistic Regression - Overall Problem

Training set: $(x^1, y^1), (x^2, y^2), (x^3, y^3), \dots, (x^m, y^m)$

Training set of m examples:
We'll use superscripts to refer to individual instances in the training set

$$y \in \{0, 1\}, \quad x \in \begin{bmatrix} x_0 \\ x_1 \\ \dots \\ x_n \end{bmatrix}, \quad x_0 = 1 \quad \text{Each example is } n+1 \text{ dimensional}$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}} = \frac{e^{\theta^T x + \theta_0}}{1 + e^{\theta^T x + \theta_0}}$$

h_{θ} is the probability of predicting the label positive ($y=+1$). Parameters: θ

Logistic Regression - Overall Problem

Training set: $(x^1, y^1), (x^2, y^2), (x^3, y^3), \dots, (x^m, y^m)$

Training set of m examples:
We'll use superscripts to refer to individual instances in the training set

$y \in \{0, 1\}$, $x \in \begin{bmatrix} x_0 \\ x_1 \\ \dots \\ x_n \end{bmatrix}$, $x_0 = 1$ Each example is $n+1$ dimensional

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}} = \frac{e^{\theta^T x + \theta_0}}{1 + e^{\theta^T x + \theta_0}}$$

h_{θ} is the probability of predicting the label positive ($y=+1$). Parameters: θ

How to choose θ ?



Logistic Regression - Cost Function

In linear regression, our cost function looks like this:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (\cancel{h(\theta)(x^i)} - y^i)^2$$

$h_{\theta}(x^i)$

This cost function works well for linear regression.

Logistic Regression - Cost Function

In linear regression, our cost function looks like this:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (h(\theta)(x^i) - y^i)^2$$

This cost function works well for linear regression.

If we use the same cost function for logistic regression, this will be a **non-convex** function of the parameters θ :

- In logistic regression, $h_{\theta}(x)$ has nonlinearity.  $h_{\theta}(x) = \frac{1}{1+e^{-\theta^T x}}$
- If you take this sigmoid function and plug it into $J(\theta)$ of linear regression, and plot $J(\theta)$:

Logistic Regression - Cost Function

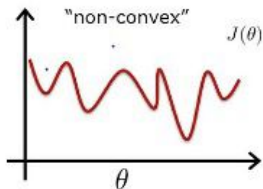
In linear regression, our cost function looks like this:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (h(\theta)(x^i) - y^i)^2$$

This cost function works well for linear regression.

If we use the same cost function for logistic regression, this will be a **non-convex** function of the parameters θ :

- In logistic regression, $h_{\theta}(x)$ has nonlinearity. $\longrightarrow h_{\theta}(x) = \frac{1}{1+e^{-\theta^T x}}$
- If you take this sigmoid function and plug it into $J(\theta)$ of linear regression, and plot $J(\theta)$:



If you run gradient descent on non-convex function, it is not guaranteed to converge to global minimum.

Logistic Regression - Cost Function

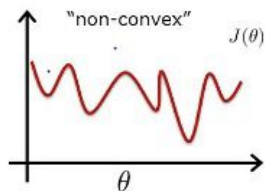
In linear regression, our cost function looks like this:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (h(\theta)(x^i) - y^i)^2$$

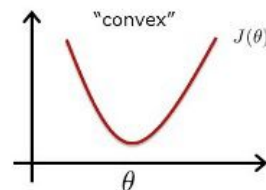
This cost function works well for linear regression.

If we use the same cost function for logistic regression, this will be a **non-convex** function of the parameters θ :

- In logistic regression, $h_{\theta}(x)$ has nonlinearity. $\longrightarrow h_{\theta}(x) = \frac{1}{1+e^{-\theta^T x}}$
- If you take this sigmoid function and plug it into $J(\theta)$ of linear regression, and plot $J(\theta)$:



If you run gradient descent on non-convex function, it is not guaranteed to converge to global minimum.



If you run gradient descent on a convex function, it will converge to global minimum.

Logistic Regression - Cost Function

Logistic regression cost function:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (h(\theta)(x^i) - y^i)^2 \quad \text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

Can we come up with a convex cost function for logistic regression? YES!



Logistic Regression - Cost Function

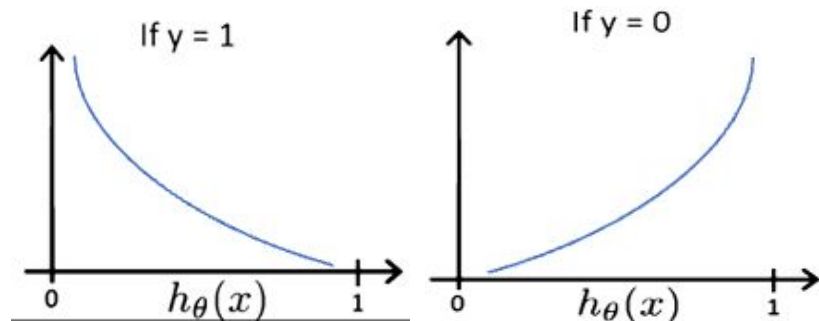
$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

If $y = 1$

- $h_{\theta}(x) = 1$, then cost is equal to 0.
- As $h_{\theta}(x) \rightarrow 0$, cost $\rightarrow \infty$ (we don't want this!)

If $y = 0$

- $h_{\theta}(x) = 0$, then cost is equal to 0.
- As $h_{\theta}(x) \rightarrow 1$, cost $\rightarrow \infty$ (we don't want this!)



Logistic Regression - Cost Function

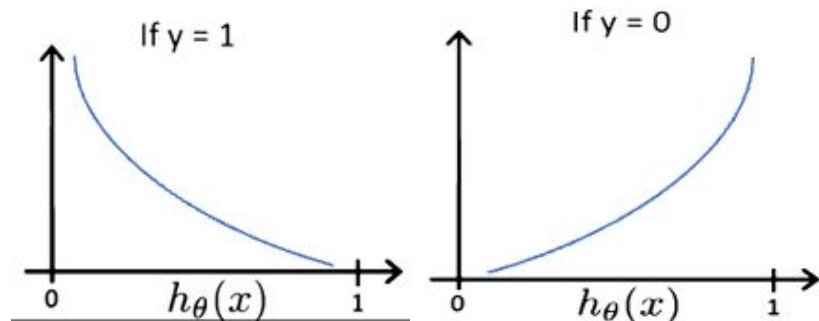
$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

If $y = 1$

- $h_{\theta}(x) = 1$, then cost is equal to 0.
- As $h_{\theta}(x) \rightarrow 0$, cost $\rightarrow \infty$ (we don't want this!)

If $y = 0$

- $h_{\theta}(x) = 0$, then cost is equal to 0.
- As $h_{\theta}(x) \rightarrow 1$, cost $\rightarrow \infty$ (we don't want this!)



Still looks too
complicate... Can
we simplify it?



Logistic Regression - Cost Function

Logistic regression cost function:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (h(\theta)(x^i) - y^i)^2 \quad \text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

Rather than 2 lines, we can compress them in 1 equation.

Logistic regression cost function:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)})$$
$$= -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

How did you see from this to this?

Why this particular cost function?

- It is **convex**! (but no closed form solution)
- It is the cost function everyone uses for logistic regression, also known as **cross-entropy** cost function.
- It can be **derived from statistics**, using maximum likelihood estimation.

Logistic Regression - Optimization

Learning/Optimization: Find parameters θ that minimizes $J(\theta)$

Gradient Descent

$$J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

Want $\min_{\theta} J(\theta)$:

Repeat {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

Compute the derivative and plug into the equation

}

(simultaneously update all θ_j)

Logistic Regression - Optimization

Learning/Optimization: Find parameters θ that minimizes $J(\theta)$

Gradient Descent

$$J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

Want $\min_{\theta} J(\theta)$:

Repeat {

$$\theta_j := \theta_j - \alpha \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

}

(simultaneously update all θ_j)

where the α come from?

A surprising fact!

The gradient descent update rule *looks identical* to linear regression, except the definition of *hypothesis* has changed!

Linear Regression: $h_{\theta}(x) = \theta^T x$

Logistic Regression: $h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$

Logistic Regression - Prediction

Prediction: Predict the output label of **a new input x** given learned parameters θ


$$\frac{p(y = 1|x)}{p(y = 0|x)} > 1?$$
$$\log \frac{p(y = 1|x)}{p(y = 0|x)} > 0?$$

If yes: positive, otherwise negative!

If yes: positive, otherwise negative!

} what's this?

Logistic Regression vs SVM

- **Logistic regression** focuses on maximizing the probability of the data. The farther the data lies from the separating hyperplane (on the correct side), the happier LR is.
- **SVM** tries to find the separating hyperplane that maximizes the distance of the closest points to the margin (the support vectors). If a point is not a support vector, it doesn't really matter.

What you should know?

- What is logistic regression? When should we use it?
- The intuition of logistic regression? what type of problems logistic regression can solve?
- Logistic regression vs SVM
- The decision boundary of logistic regression
- How to perform learning and prediction under logistic regression?

Suggestion: If you're not very clear or want to learn more, please do read: "C. Bishop: Pattern Recognition and Machine Learning. Springer, 2006" (recommended text book). It will help you to understand the concept.

Please study the lecture notes



Acknowledgement

- *The following material have been referenced or partially used:*
 - *Lecture notes from Prof. Berrak Sisman and Prof. Zhao Na*
 - *National University of Singapore – Pattern Recognition Module (EE5907R)*
 - *National University of Singapore – Neural Networks Module (EE5904R)*
 - *University of Edinburgh, United Kingdom – Machine Learning And Pattern Recognition (MLPR, INFR11130)*
 - *Stanford University – Machine Learning (CS229)*